Deterministic Optimization

Linear Programming Duality

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Introduction to Duality Theory

LP Duality

Learning Objectives for this lesson

- Recognize the motivation of duality
- Examine the mechanism of duality

Linear Programming Duality

- LP duality is at the **core** of linear programming theory
- It gives us a new perspective on understanding LP, is important for designing algorithms, and has many applications (pricing, game theory, robust optimization, and many more)
- LP duality has deep connections to a more general principle, called Lagrangian relaxation, which is applicable to a wide range of optimization problems

Motivation of LP Dual

- A fundamental motivation of LP duality is to find a systematic way to construct a *lower bound* to the original LP.
- Original LP (Primal LP): $Z_P = \min\{c^\top x : Ax = b, x \geq 0\}$
- Any feasible solution x provides an *upper bound* on Z_P, ie $Z_P \leq c^\top x$
- What about a lower bound, i.e. $Z_D \leq Z_P$?
- This lower bound is useful, because if the lower bound is very close to an upper bound, we have a good estimate of the true optimal.
- However, to get a lower bound, we need to *modify* the original LP.
- In particular, we need to *relax* the problem.

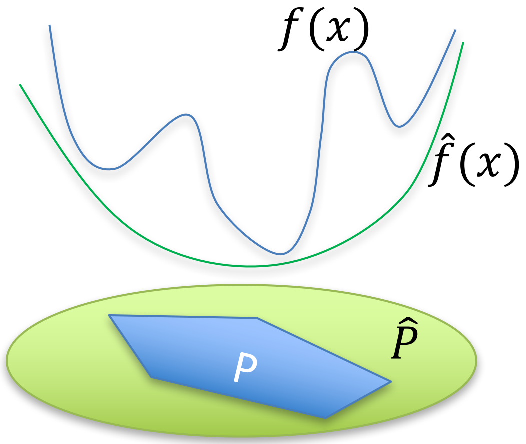
Principle of Relaxation

- What do we mean by relaxation?
  - First, find a new objective function that is always smaller or equal to the original objective function at any feasible point
  - Second, find a feasible region that is larger than the feasible region of the original problem
- Combining the two steps:
  - Minimize a function *lower* than the original objective over a region that is *larger* than the original one
  - The optimal objective of the new problem will be a *lower bound* to the original one
- Among all possible relaxations and lower bounds, find the *best* lower bound

Illustration

- Original problem
  - $\min f(x)$
  - s.t. $x \in P$

- Relaxed problem
  - $\min \hat{f}(x)$
  - s.t. $x \in \hat{P}$



- Principles of relaxation works for general optimization problems

Summary

- We learned:
  - Motivation behind LP duality
  - Principle of relaxation applicable to a wide range of optimization problems, far beyond LP